ISM-Band SDR Board

1 Goal

I'm building a receive-only software defined radio for the US 902 MHz to 928 MHz ISM band. The board takes a signal from an antenna, filters and amplifies it, mixes it down to I and Q at audio frequencies, and hands it to a USB audio interface. The PC does the rest in software.

2 Outline

Everything is on one 4-layer board, drawn as four Altium sheets:

- **Front end:** SAW filter, low-noise amplifier, second SAW filter.
- **LO:** a PLL synthesizer locked to a TCXO, and the STM32 that programs it.
- **Tuner:** the I/Q demodulator and the baseband amplifier and filter for each channel.
- **Power:** USB-C 5 V, a ferrite bead per branch, and LDOs for the 3.3 V rails.

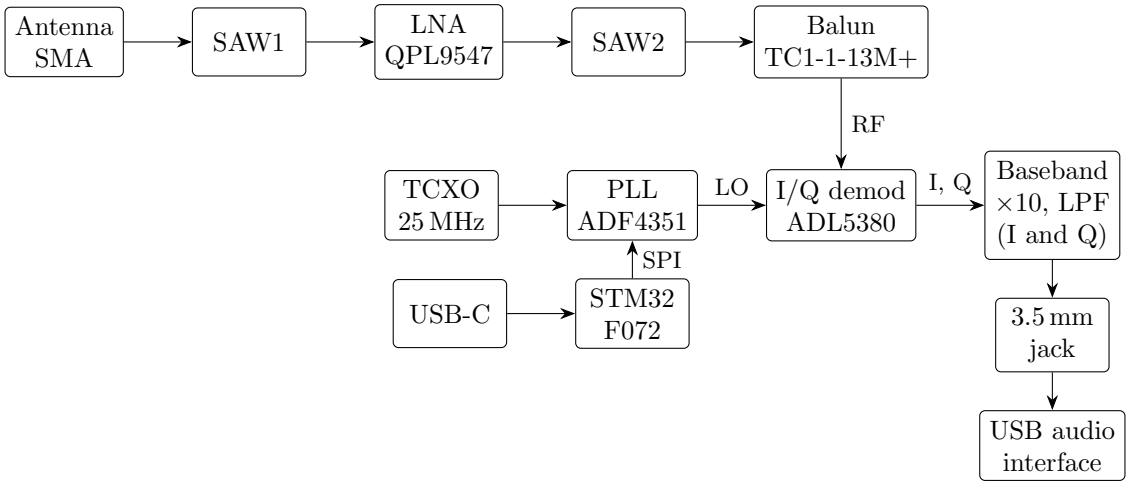
The I and Q outputs go to a 192 kS/s USB audio interface, so the board sees a 192 kHz window of spectrum at a time. The STM32 tunes the LO over USB.

3 Components

Main parts only. Resistors, capacitors, inductors and ferrite beads are in the BOM.

Part	Function	Qty
Abracon AFS20A26-915.00	915 MHz SAW band-pass filter	2
Qorvo QPL9547	Low-noise amplifier	1
Mini-Circuits TC1-1-13M+	RF balun (single-ended to differential)	1
Analog Devices ADL5380	I/Q demodulator	1
Analog Devices ADF4351	PLL synthesizer (LO)	1
25 MHz TCXO, ±0.5 ppm	LO reference	1
Analog Devices LT6203	Dual op-amp, baseband (one per channel)	2
ST STM32F072CBT6	Controller, USB	1
TI TPS7A2033	3.3 V LDO (LO rail, digital rail)	2
ST USBLC6-2SC6	USB ESD protection	1
Edge-launch SMA	Antenna input	1
GCT USB4105-GF-A	USB-C, power and data	1
CUI SJ2-357X	3.5 mm jack, I and Q out	1

4 Block diagram



USB-C also supplies 5 V to the whole board: the LNA, ADL5380 and op-amps run from 5 V behind their own ferrite beads, and the ADF4351/TCXO and the STM32 each get their own TPS7A2033.

5 Signal chain

5.1 Gain and noise figure

Values are typical datasheet numbers at 915 MHz; the trace and connector loss is an estimate. Cumulative noise figure is from the Friis formula,

$$F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \cdots$$

with F and G as linear ratios.

Stage	Gain (dB)	NF (dB)	Total gain (dB)	Total NF (dB)
SAW1 (AFS20A26)	−1.7	1.7	−1.7	1.70
LNA (QPL9547)	+24.6	0.38	22.9	2.08
SAW2 (AFS20A26)	−1.7	1.7	21.2	2.09
Traces and connectors	−0.3	0.3	20.9	2.09
Balun (TC1-1-13M+)	−0.4	0.4	20.5	2.09
I/Q demod (ADL5380)	≈ +6	≈ 11	26.5	2.36
Baseband amplifier	+20 (voltage)	—	—	—

The system noise figure is about 2.4 dB, and almost all of it is set before the LNA. After about 21 dB of gain, the ADL5380's 11 dB noise figure adds less than 0.3 dB, and the baseband stages add effectively nothing.

5.2 Following one signal through the board

As an example, take a sensor transmitting FSK at 915.000 MHz, about 50 kHz wide, arriving at −90 dBm. Thermal noise is −174 dBm/Hz, so in 50 kHz the noise floor at the antenna is $-174 + 10\log_{10}(50\,000) = -127$ dBm, and the signal starts with 37 dB of SNR. I tune the LO to 914.950 MHz, 50 kHz below the signal.

Antenna and SAW1. The antenna hears the wanted signal together with everything else, including the cell downlink at 869 MHz to 894 MHz, which near a tower can be 60 dB stronger than my signal. SAW1 passes 902 MHz to 928 MHz with 1.7 dB loss and cuts the cell band by at least 40 dB, so the LNA never sees it at full strength. A 47 nH shunt inductor at the SMA drains static to ground and is invisible at 915 MHz. Because SAW1 is passive, its noise figure equals its loss: the signal is now at -91.7 dBm and the SNR drops to 35.3 dB. This is the price of protecting the LNA.

LNA. The QPL9547 adds 24.6 dB of gain with a 0.38 dB noise figure, so the signal is at about -67 dBm. Signal and noise rise together, so the SNR barely moves. This is the most important stage: once the signal is this large, the noise added by everything after it hardly counts. The bias circuit follows Qorvo's evaluation board, and the part draws about 65 mA from 5 V.

SAW2 and traces. The second SAW removes the out-of-band noise and signals that the LNA amplified, so they can't mix into the output later. With the trace and connector loss, the front end totals 20.9 dB of gain and about 2.1 dB of noise figure. The signal leaves it at about -69 dBm, roughly 80 µV rms in 50 Ω. All RF traces are 50 Ω grounded coplanar waveguide over a solid ground plane.

Balun. The TC1-1-13M+ turns the single-ended signal into the differential pair the ADL5380's RF inputs expect, through series DC-blocking capacitors, at the cost of about 0.4 dB.

LO. The ADF4351 generates the LO from the 25 MHz TCXO. The TCXO is accurate to ±0.5 ppm, about ±460 Hz at 915 MHz, and the LO inherits that accuracy. The STM32 receives a frequency from the PC over USB, calculates the PLL registers and writes them over SPI, then watches the lock-detect pin. The ADF4351's differential output goes through a small low-pass filter into the ADL5380's LO inputs.

I/Q demodulator. Inside the ADL5380 the signal is multiplied by the LO twice, once by a cosine to make I and once by a sine to make Q. Multiplying two sine waves gives their difference and their sum: the difference, $915.000 - 914.950 = 50$ kHz, is what I keep; the sum, near 1830 MHz, is far above anything the baseband passes. Every signal in the window shifts the same way, so 915.020 MHz comes out at 70 kHz and 914.900 MHz at -50 kHz.

With I alone, signals at 50 kHz and -50 kHz would look identical. With both I and Q, software can tell which side of the LO each one is on, which makes the full 192 kHz window usable. The conversion gain is about 6 dB, so each output carries about 160 µV rms. Its 11 dB noise figure only raises the system noise figure from 2.1 dB to about 2.4 dB.

Placing the signal 50 kHz away from the LO keeps it off 0 Hz, where LO leakage mixing with itself produces a DC offset. Any mismatch between I and Q leaves a faint mirror image of the signal at -50 kHz, which software corrects.

Baseband. Each channel has one LT6203, running from 5 V around a 2.5 V mid-rail reference. The ADL5380's outputs are AC-coupled through 10 µF capacitors, which blocks its DC level. The first op-amp is a difference amplifier with a gain of 10 (20 dB). It turns each differential pair into a single signal, cancels noise picked up equally on both lines, and brings the signal to about 1.6 mV rms. A capacitor across its feedback resistor puts a pole at about 80 kHz. The second op-amp

is a unity-gain Sallen-Key stage at about 80 kHz with $Q \approx 1$. Together the two stages form a 3rd-order Butterworth low-pass at 80 kHz. It keeps the audio interface from being overloaded by strong signals outside the window.

Output. Each channel leaves through a 100 Ω series resistor, which keeps the op-amp stable into cable capacitance, then a 10 µF capacitor that removes the 2.5 V offset, with 100 kΩ to ground. I goes to the tip and Q to the ring of one 3.5 mm stereo jack.

Audio interface and PC. The interface samples I and Q at 192 kS/s each. Treated as one complex signal $I + jQ$, that is a spectrum from −96 kHz to 96 kHz around the LO. In software I correct the I/Q imbalance, shift the signal from 50 kHz to 0 Hz, low-pass filter it to its own bandwidth, decimate, and demodulate. With a 2.4 dB system noise figure, the −90 dBm example signal ends up with about 34.6 dB of SNR, close to the 37 dB it had at the antenna.